

STREAMING DATA ANALYTICS

Production-Grade Pipeline & Dashboard

E-commerce Growth & Retention Command Centre

Project	Details
Student	Parv Jhamb
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Programme	PGDM, Batch-27 — FORE School of Management, New Delhi
Specialization	Big Data Analytics and Marketing
Course	Streaming Data Analytics
Project Scope	Production-grade extension of Assignments 1–3
Core Stack	Kafka • Zookeeper • MySQL • Grafana • Python • FastAPI • Docker
Business Lens	E-commerce growth, conversion, retention, campaign performance and anomaly response
Submitted to	Prof. Aditya Dua

Executive framing

This report documents the transformation of an academic streaming data pipeline into a production-ready decision-support platform. The emphasis is not only on moving events through Kafka but also on reliability, observability, anomaly response, customer intelligence, API access, and executive decision-making.

1. Executive Summary

The project extends the original Streaming Data Analytics assignment from a basic Kafka producer/consumer and Grafana dashboard into a production-grade, containerised e-commerce analytics platform. The system combines real-time event streaming, 15-second analytical windows, MySQL persistence, statistical anomaly detection, RFM customer segmentation, a FastAPI service layer, and a rebuilt CMO-level Grafana dashboard.

The platform is designed around a simple executive question: “What is happening in the business right now, why does it matter, and what should a growth or retention team investigate next?” Rather than treating Grafana as the data source, the architecture separates event generation, streaming computation, persistence, APIs and visualisation so that multiple downstream consumers can use the same analytical layer.

Capability	Business value	Implementation
Real-time streaming	Near-real-time view of commercial activity	Kafka + Python producers/consumers
Reliability	Prevents silent data loss and isolates bad messages	DLQ + manual offset commits + retry/backoff
Anomaly detection	Detects unusual conversion, abandonment, revenue, and spend behaviour	Rolling statistical baseline
Customer intelligence	Moves from aggregate KPIs to individual retention segments	RFM quintile scoring
Decision access	Makes data consumable beyond Grafana	FastAPI REST layer with 8 endpoints
Executive dashboard	Converts streaming data into a CMO operating view	Grafana dashboard with filters, alerts and business panels

Executive takeaways

- The dashboard connects acquisition, funnel performance, campaign activity and retention into one operating view.
- The RFM layer identifies customer groups that can be acted on differently instead of treating the entire customer base uniformly.
- The alerting layer turns statistical deviations into operational signals and sends critical notifications through Grafana's alerting pipeline.
- The system was deliberately tested against failure and edge cases; several implementation defects were discovered and fixed before the final deployment.
- The figures shown in the dashboard are synthetic demonstration data. They validate the pipeline mechanics and analytical workflow, not real company performance.

Overall, the project demonstrates an end-to-end data product mindset: event generation → streaming → resilient processing → analytical persistence → customer intelligence → API exposure → executive visualisation → alert-driven action.

2. Business Problem & Objectives

A conventional dashboard built from batch data can describe what happened. Still, a streaming analytics platform is expected to shorten the time between an event, its detection and a business response. The project, therefore, treats the e-commerce operation as a continuously changing system.

Business question	Dashboard/system answer
Are customers converting efficiently?	Conversion and cart-abandonment KPIs plus funnel view
Where are customers dropping out?	Views → Cart Adds → Purchases funnel.
Which customer groups need attention?	RFM segment distribution and at-risk customer list
Are metrics behaving unusually?	Statistical anomaly detection and alert annotations
Which campaigns generate engagement?	Campaign click/spend/CTR view and campaign → segment breakdown.
Where is customer value concentrated?	Segment value table and lifetime-value view
What are customers searching for?	Top search query panel
Is traffic intensity changing?	Time-based traffic heatmap

2.1 Project objectives

1. Containerise the complete streaming stack so the project can be started consistently.
2. Make the consumer resilient to malformed events, transient Kafka failures and database outages.
3. Tie Kafka offset commits to successful database writes to avoid silent data loss.
4. Add anomaly detection based on each consumer instance's recent statistical behaviour.
5. Capture customer-level purchase data and build an RFM segmentation layer.
6. Expose metrics, alerts and segmentation data through a REST API.
7. Rebuild the dashboard around executive questions rather than isolated technical charts.
8. Validate the system through simulations, exhaustive input sweeps, integration tests and live deployment checks.

3. Solution Architecture

The final design follows a layered architecture. Event producers generate e-commerce activity and publish it to Kafka. Consumers process events into analytical windows and customer-level records, while MySQL provides durable storage. RFM segmentation operates over transaction history. FastAPI exposes the resulting information to external consumers, while Grafana acts as the executive visualisation and alerting layer.

Layer	Primary components	Responsibility
Event generation	Python producers	Generate clickstream, cart, transaction, advertising and customer activity events.
Streaming backbone	Zookeeper + Kafka	Transport, buffer and decouple streaming

		events
Processing	Python consumer services	Validate, aggregate, calculate KPIs and detect anomalies
Reliability	DLQ + retry/backoff + manual commits	Prevent consumer crashes and reduce data-loss risk
Storage	MySQL + named Docker volumes	Persist analytical windows, alerts and customer transactions
Customer intelligence	RFM segmentation service	Score customers and classify them into eight segments
API	FastAPI	Expose metrics, alerts and segment information via 8 endpoints
Visualization	Grafana	Executive dashboard, filters, annotations and alerting

3.1 Data flow

9. Synthetic e-commerce events are produced continuously.
10. Kafka receives and partitions the events by topic.
11. Consumers validate messages and aggregate business metrics over 15-second tumbling windows.
12. Successful writes are persisted to MySQL; malformed/unprocessable messages are redirected to a dead-letter topic.
13. Recent history is used to establish statistical baselines for anomaly detection.
14. Customer transaction records feed the RFM segmentation service.
15. FastAPI exposes the resulting analytical data for programmatic consumption.
16. Grafana queries MySQL/API-backed data to present the executive command centre and trigger alerts.

4. Engineering Evolution: Six Production Phases

Phase 0 — Foundation Hardening

Manual execution of `producer.py` and `consumer.py` was converted into Docker services. Structured logging, environment-based configuration, retry/backoff logic and a Docker Compose stack were introduced. Named volumes protect persisted data from container recreation, while schema initialisation automates first-start database setup.

Phase 1 — Reliability & Scalability

A dead-letter queue isolates malformed messages; `consumer_id` makes parallel consumers distinguishable; and manual Kafka offset commits are tied to successful MySQL writes. A real offset-ordering defect was found during testing and corrected, so a database failure results in reprocessing rather than a silent, permanent loss.

Phase 2 — Alerting

Fixed thresholds were replaced by statistical anomaly detection. Conversion rate, cart abandonment rate, revenue and ad spend are compared with the recent history of the same consumer instance. Alerts are stored and surfaced through Grafana, with critical alerts capable of sending email via SMTP.

Phase 3 — RFM Customer Segmentation

The pipeline was extended with customer_transactions and an RFM scoring service. Customers are scored on Recency, Frequency and Monetary value using quintiles and classified into eight business segments. Exhaustive testing of all 125 possible score combinations exposed and fixed a rule-ordering issue, while recency was moved to minute-level granularity for the short-duration demo environment.

Phase 4 — REST API

FastAPI was introduced as a formal service layer. Eight endpoints expose metrics, alerts and segmentation data with interactive documentation. The client-supplied endpoint was tested against a deliberate SQL-injection attempt and rejected cleanly with HTTP 400.

Phase 5 & 6 — Dashboard Rebuild & Advanced Analytics

The Grafana dashboard was rebuilt into executive sections covering summary KPIs, funnel and segments, trends, category/campaign performance, operational pulse, segment value, search behaviour and traffic intensity. New data capture enabled segment-level trend analysis and campaign-to-segment analysis.

5. Executive Dashboard Walkthrough

The dashboard is intentionally organised as an operating narrative rather than a collection of unrelated visualisations. The sequence moves from “what is happening?” to “where is the issue?” to “which customers/campaigns are involved?” and finally to “what can the team act on?”

3.1 Executive Summary

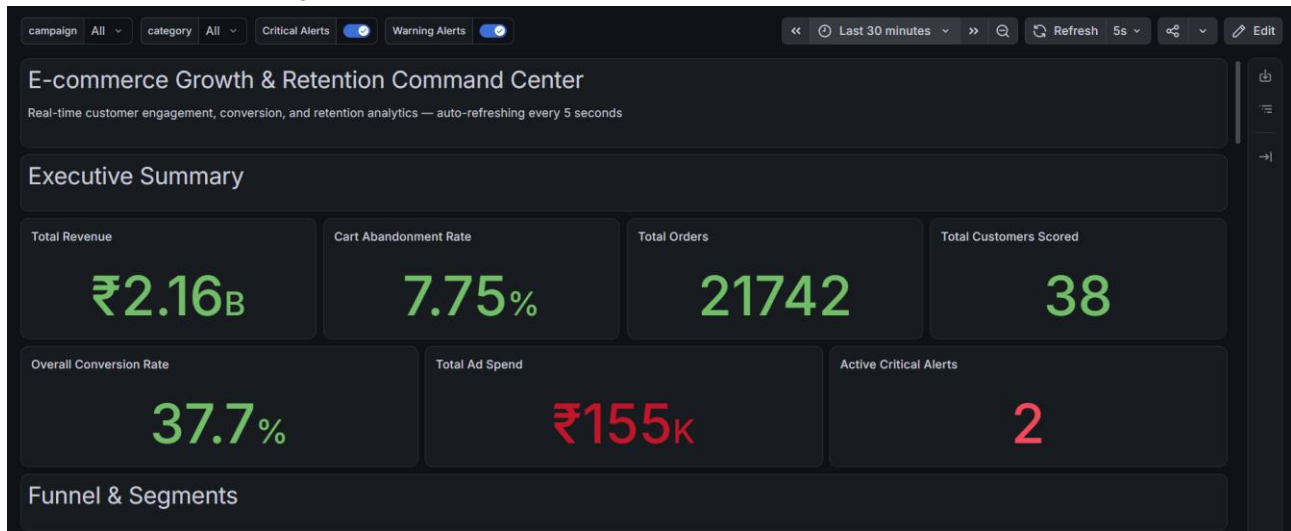


Figure 1. Executive Summary — top-level KPI strip, filters and alert controls.

The executive strip reports Total Revenue, Cart Abandonment Rate, Total Orders, Total Customers Scored, Overall Conversion Rate, Total Ad Spend and Active Critical Alerts. Campaign and category filters drive the dimension-aware panels. Rate-based KPIs remain intentionally unfiltered because the underlying pipeline does not attribute individual conversions to a specific campaign/category.

3.2 Funnel & Customer Segments

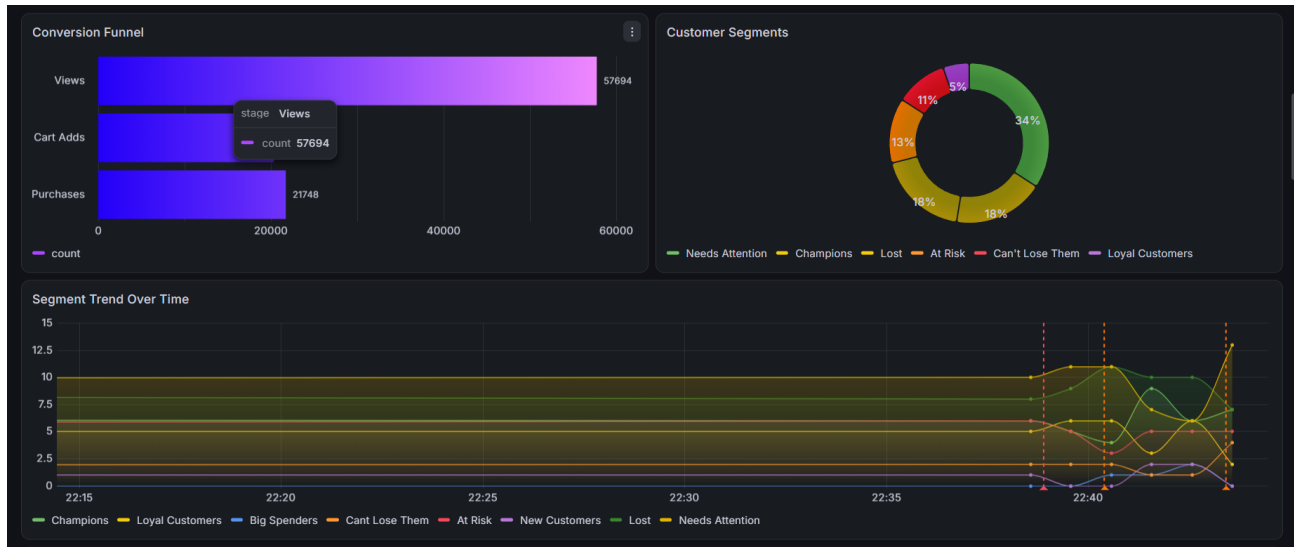


Figure 2. Funnel and RFM segment distribution.

The funnel moves from 57,694 views to 21,748 purchases, while the segment doughnut shows the composition of the scored customer base. The panel is designed to connect acquisition/funnel leakage with retention structure.

3.3 Segment Trend & Campaign → Segment

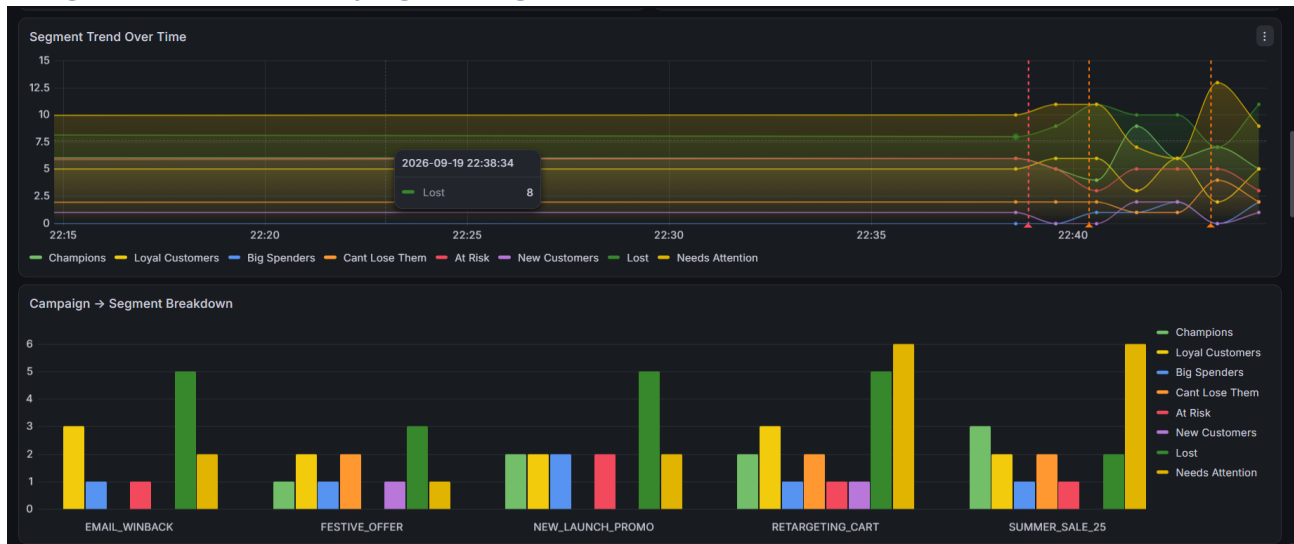


Figure 3. Segment movement over time and campaign-to-segment breakdown.

The apparent flat period followed by divergence corresponds to the documented database reset/restart event. After fresh activity resumes, segment populations separate. The campaign-to-segment view adds a dimension that ordinary campaign performance tables lack: it shows which customer-value states campaign engagement reaches.

3.4 Trends

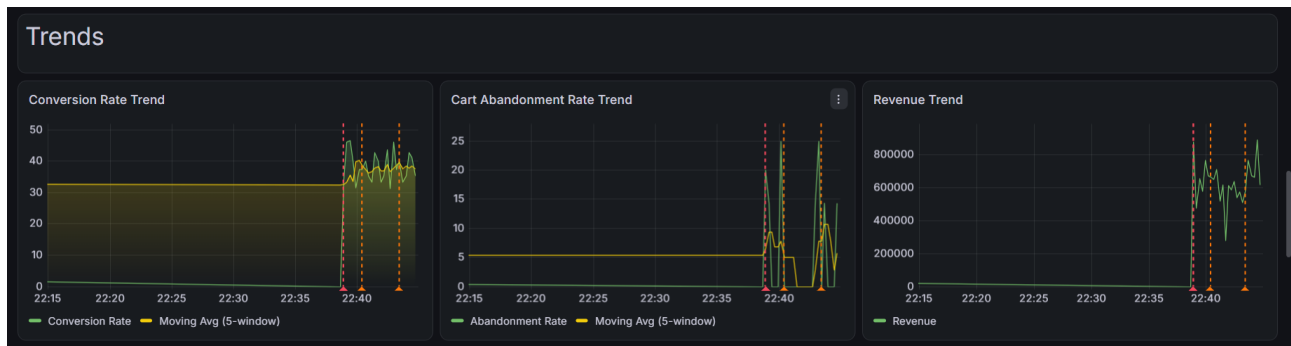


Figure 4. Conversion, abandonment and revenue trends with anomaly annotations.

The trend panels combine live values, five-window moving averages and anomaly annotations. Red/orange vertical markers represent critical/warning events identified by the anomaly engine, allowing a viewer to compare the signal with the underlying metric movement.

3.5 Category & Campaign Performance

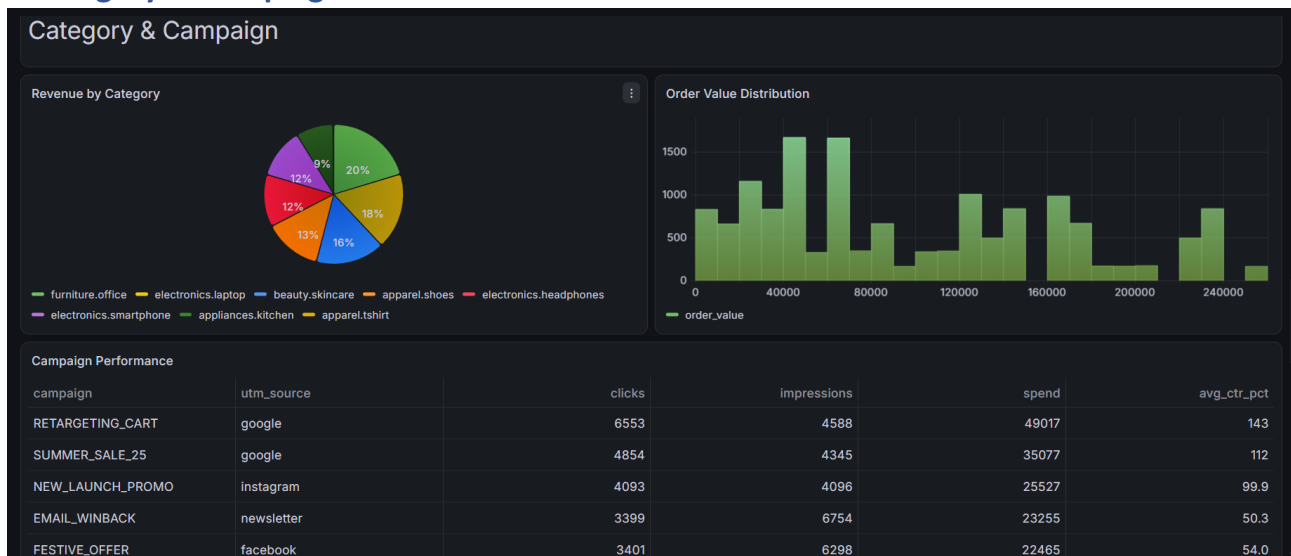


Figure 5. Revenue by category, order-value distribution and campaign performance.

The category chart shows a diversified revenue mix; the histogram highlights multiple order-value clusters; and the campaign table provides clicks, impressions, spend and CTR. The campaign panel is intended for performance diagnosis rather than treating spend alone as a success/failure indicator.

3.6 Operational Pulse

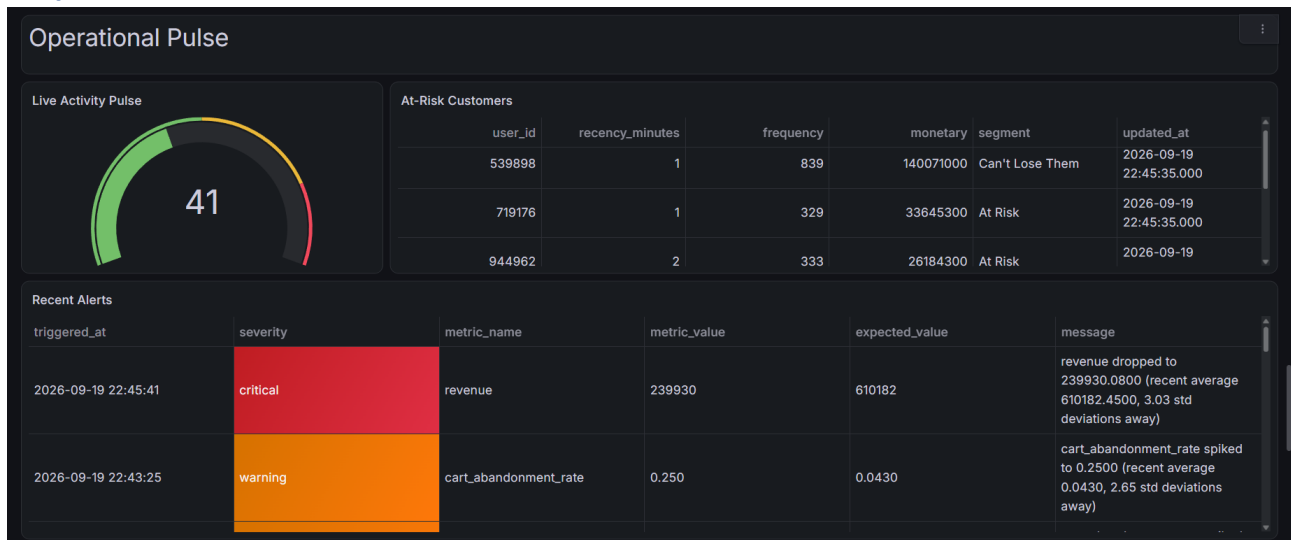


Figure 6. Live activity, at-risk customer list and alert log.

The live activity gauge provides a quick operational heartbeat. The at-risk table connects recency, frequency and monetary history to specific customer IDs, while the alert table provides a severity-coded operational log.

3.7 Segment Value, Search & Traffic

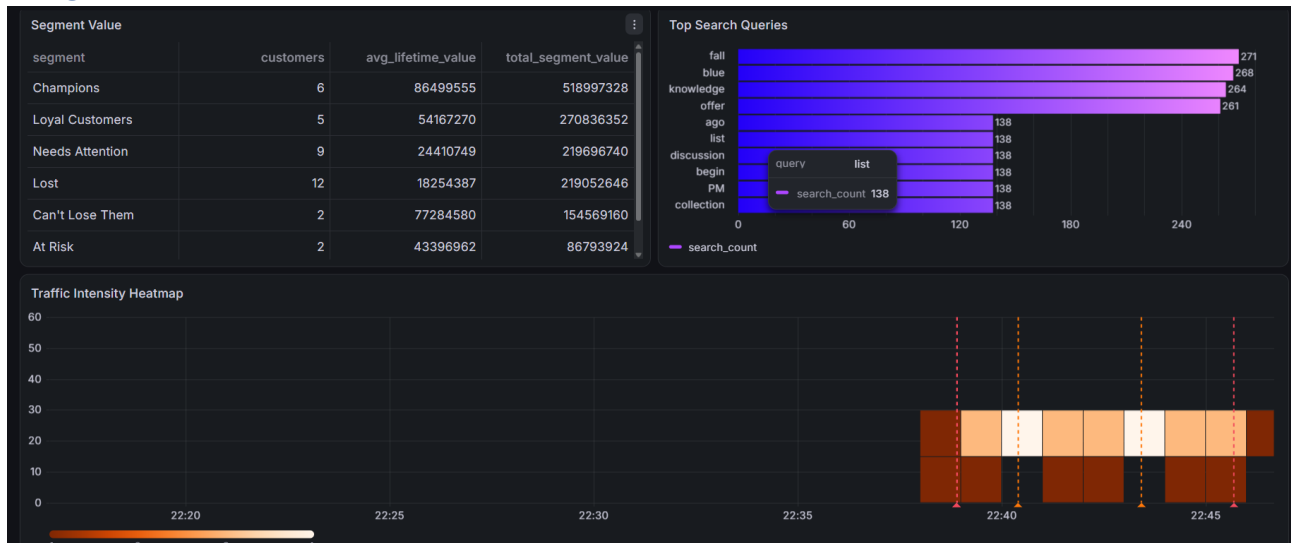


Figure 7. Segment value, top search queries and traffic intensity heatmap.

The segment-value view translates customer segmentation into monetary terms. The search panel exposes demand signals, and the heatmap shows how traffic intensity changes over the streaming run.

6. Key Business Insights

- The 57,694-to-21,748 funnel indicates a large gap between browsing and purchasing; improving on-site engagement and cart progression is therefore a material business lever in the simulated environment.

- The segment distribution shows that a substantial share of customers is outside the clearly loyal group, creating a strong retention and reactivation use case for RFM-driven targeting.
- Six Champions account for approximately ₹51.9 crore of lifetime value in the captured snapshot, illustrating how a small high-value segment can warrant differentiated retention treatment.
- Lost and Needs Attention customers still represent significant historical value, so disengagement does not imply low value. This supports a win-back lens rather than treating these users as equivalent to never-converted prospects.
- RETARGETING_CART reaches visible Champions and Needs Attention populations, consistent with its role as a re-engagement mechanism.
- NEW_LAUNCH_PROMO reaches a visible At Risk population in the captured data, creating a useful question for campaign targeting and audience definition.
- Cross-metric anomaly events demonstrate the value of streaming analytics: conversion, abandonment and revenue can be assessed together rather than as isolated dashboard tiles.

7. Testing, Reliability & Controls

Control/test	What was validated	Outcome
9-window anomaly simulation	8 stable windows followed by 1 deliberately broken window	Stable periods produced zero false alarms; broken windows triggered alerts
RFM exhaustive sweep	All 125 possible R/F/M score combinations	Rule-ordering issue identified and corrected
Recency granularity test	Minute-level vs day-level recency in a short demo	Minute-level granularity retained discrimination
Offset/DB failure reasoning	Commit only after a successful MySQL write	Prevents silent loss and enables re-processing
Dead-letter handling	Malformed/unprocessable messages	Isolates failures without crashing the consumer
API security test	Deliberate SQL-injection attempt on client-supplied endpoint	Rejected with a clean HTTP 400
Dashboard validation	Filters, annotations, KPIs and operational panels	Validated against captured live data and expected interactions

8.1 Reliability principles

- Fail safely: bad messages are isolated rather than allowed to terminate the stream processor.
- Persist before acknowledging: Kafka offsets are committed only after the downstream database operation succeeds.
- Observe continuously: structured logs and Grafana alerting make operational issues visible.
- Test before deployment: simulations and exhaustive input sweeps were used to find defects that ordinary happy-path testing would miss.
- Separate concerns: streaming, storage, segmentation, API and visualisation are independent layers.

8. API & Integration Layer

The FastAPI layer establishes the platform as a reusable analytical service rather than a dashboard-only application. Eight REST endpoints expose metrics, alerts and segmentation information and provide interactive documentation. This enables future consumers, such as a React/Next.js front end, internal operations tooling, or additional BI surfaces, to use the same analytical foundation.

Integration capability	Strategic implication
REST metrics access	Allows analytics to be consumed outside Grafana
Alert retrieval	Supports operational workflows and future notification systems
Segment access	Enables CRM/retention tooling to consume customer intelligence
Interactive API documentation	Improves developer usability and integration speed
Input validation	Reduces risk from malformed or malicious client inputs

9. Data & Repository Design

The system stores both aggregate analytical windows and customer-level transaction records. This is a deliberate design choice: aggregate windows support fast executive metrics, while customer_transactions supports retention intelligence that cannot be reconstructed from aggregate totals alone.

Area	Representative content
Kafka topics	Event streams for clickstream, cart, transaction and advertising activity
Analytical windows	Conversion rate, cart abandonment rate, revenue, ad spend and event volume
Alerts	Metric, severity, expected value, observed value and message/context
Customer transactions	Customer purchase history used by RFM
RFM segments	Champions, Loyal Customers, Big Spenders, Can't Lose Them, At Risk, New Customers, Lost, Needs Attention
Configuration	Environment variables and Docker Compose service configuration

10. Data Limitations & Responsible Interpretation

- All business data in the demonstration is synthetic/Faker-generated and replayed through Kafka. Monetary totals, customer counts and percentages are illustrative rather than real business results.

- The captured dashboard run contains a CTR realism issue caused by the original synthetic event-generation logic. The visualization is correct for the captured data; the upstream generator has a tested fix that constrains clicks to be a subset of impressions.
- The captured search query panel contains generic dictionary-style terms because the earlier generator did not create realistic product search phrases. A tested generator improvement is available for future regenerated data.
- The demo compresses customer behaviour that would normally unfold over weeks or months into minutes. This is why 15-second windows and minute-level RFM recency are appropriate for demonstrating the mechanics.
- The captured RFM snapshot contains 38 scored customers. This is sufficient to validate the mechanics and dashboard flow, but not to make statistically representative claims about a real customer population.
- The database restart/reset visible in several trend panels is part of the captured run and should be read as an operational event rather than a rendering defect.

11. Roadmap & Next Steps

Priority	Next step	Business impact
P0	Regenerate the producer dataset using the corrected CTR and realistic search-query generators.	Improves the semantic realism of demo outputs and removes avoidable data-quality confusion.
P0	Persist historical data across longer demo runs and avoid destructive resets.	Makes trend and RFM movement more representative.
P1	Introduce schema/data-quality contracts for every event type.	Prevents invalid business relationships from reaching analytical tables.
P1	Move from synthetic data to a controlled real/sanitised event source if available.	Enables real business validation and meaningful KPI benchmarking.
P1	Add role-based API authentication and secret management for deployment outside a classroom environment.	Strengthens production security.
P2	Add campaign ROI and customer-level attribution where the event model supports it.	Moves from descriptive campaign reporting toward investment optimisation.
P2	Add automated CI/CD and container health checks.	Improves repeatability and operational governance.
P2	Add a front-end layer consuming the FastAPI service.	Separates presentation from Grafana and enables a broader executive product experience.

11.1 Target operating model

In a real organisation, the platform can evolve into a Growth & Retention Control Tower: Kafka becomes the event backbone; the streaming layer computes operational KPIs; RFM and future predictive models create customer intelligence; FastAPI exposes the analytical services; and Grafana or a dedicated web application provides executive and operational interfaces. Alerts can then become workflow triggers rather than simply dashboard annotations.

12. Conclusion

The project demonstrates the progression from an academic streaming assignment to a production-shaped analytics platform. Its value lies not only in the number of technologies used but also in the architectural decisions that connect them: durable streaming, failure isolation, correct offset semantics, statistical monitoring, customer-level segmentation, API abstraction, and decision-oriented visualisation.

The implementation also demonstrates an engineering mindset based on finding defects before stakeholders do. The project uncovered an offset-commit/data-loss risk, an RFM rule-ordering defect, a recency-granularity issue and synthetic data-generation realism problems. These were treated as engineering findings, tested and corrected or explicitly documented.

Final assessment

The delivered system is best described as production-shaped and production-minded: the architecture, controls, and analytical workflow mirror a real deployment pattern, while the synthetic data, small sample size, and classroom-scale runtime remain the primary constraints on business realism.

Appendix A — Dashboard Screenshot Index

Figure	Section	Purpose
Figure 1	Executive Summary	CMO-level KPI strip and filters
Figure 2	Funnel & Customer Segments	Conversion funnel and RFM mix
Figure 3	Segment Trend & Campaign → Segment	Retention movement and audience reach
Figure 4	Trends	Conversion, abandonment and revenue with alerts
Figure 5	Category & Campaign	Category mix, order values and campaign metrics
Figure 6	Operational Pulse	Live activity, at-risk customers and alert log
Figure 7	Segment Value / Search / Heatmap	Value concentration, search demand and traffic intensity